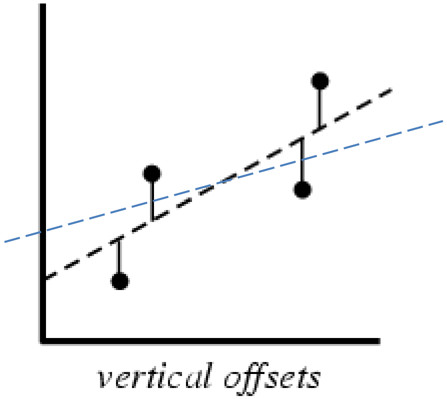
ISIE-SEM Summer School
Bath, 03.07.2026

Dealing with Uncertainties in Material Flow Analysis

Oliver CENCIC
Institute for Water Quality and Resource Management
TU Wien

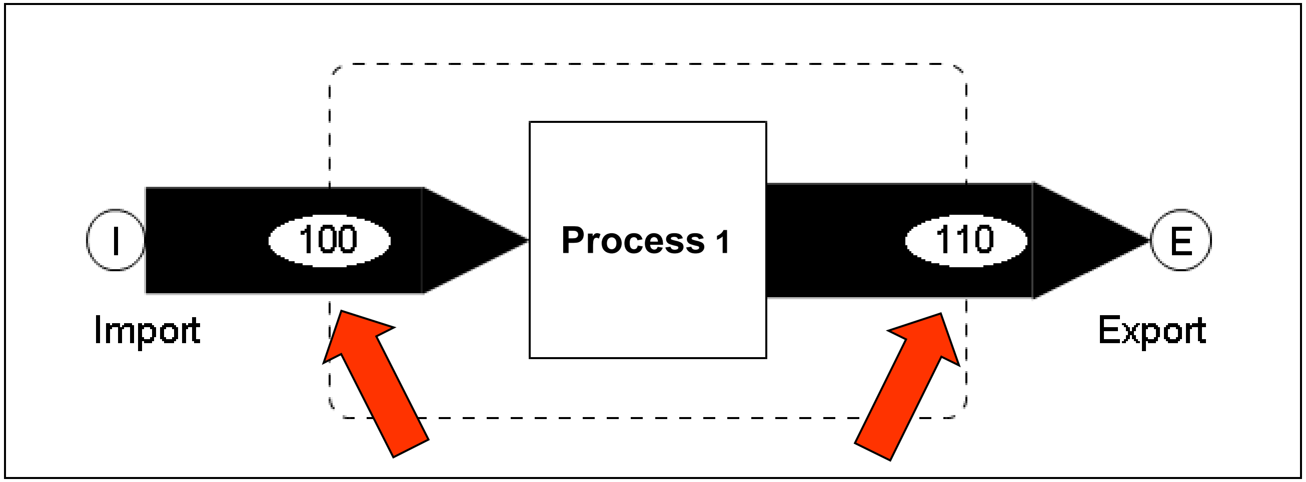
C. F. Gauss
(1777-1855)

Method of least squares



→ **Basis of Data Reconciliation**

Data Reconciliation (2)



contradictional data!
(Import ≠ Export)

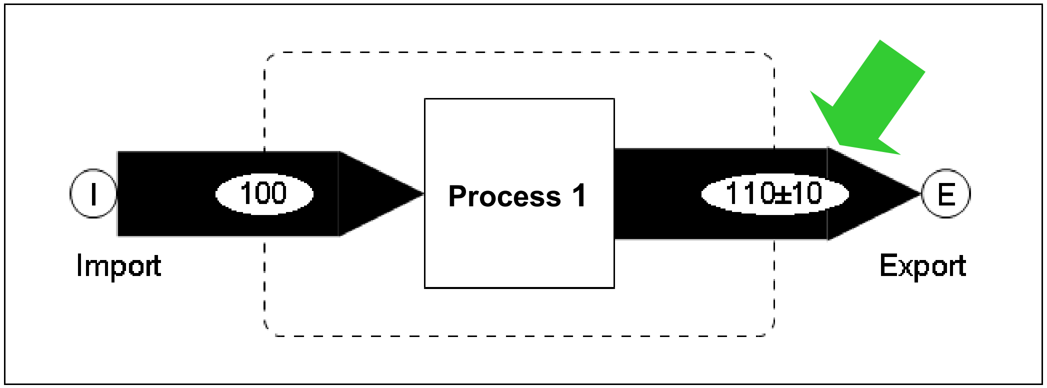
BUT:

1 balance equation, 0 unknown variables

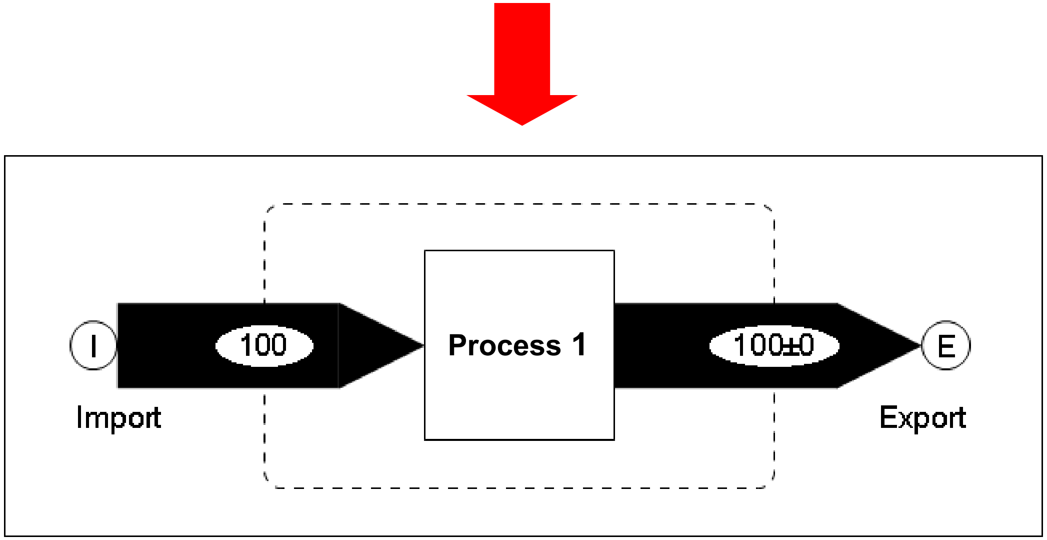
→ „over determined“ system of equations
→ data uncertainties given

Data Reconciliation possible!

Data Reconciliation (3)



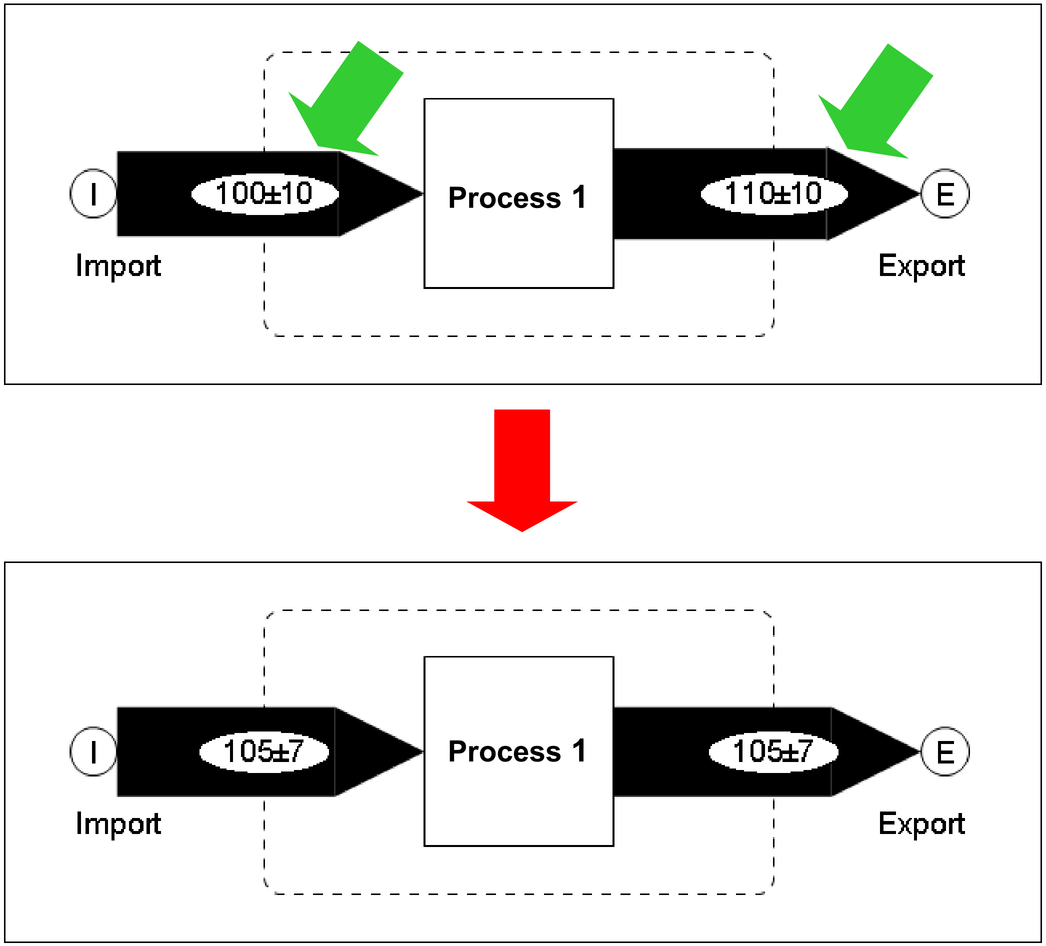
only export flow uncertainty given



→ contradiction resolved
→ uncertainty reduced

values without uncertainties are not reconciled!

Data Reconciliation (4)



import and export flow have the same absolute uncertainty

→ contradiction resolved
→ uncertainty reduced

values with the same absolute uncertainty are changed to the same extent!

Data Reconciliation (5)

The result of the data reconciliation procedure in this simple case is the weighted mean:

$$\bar{x} = \frac{\bar{x}_1 \cdot w_1 + \bar{x}_2 \cdot w_2}{w_1 + w_2} \quad with \quad w_1 = \frac{1}{{s_{\bar{x}_1}}^2}, w_2 = \frac{1}{{s_{\bar{x}_2}}^2}$$

Calculation of uncertainty with error propagation:

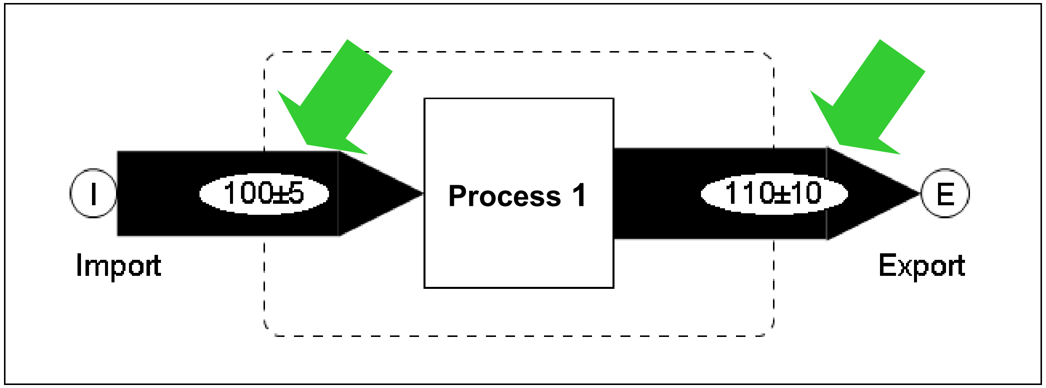
$$ {s_{\bar{x}}}^2 = \left(\frac{w_1}{w_1 + w_2}\right)^2 \cdot {s_{\bar{x}_1}}^2 + \left(\frac{w_2}{w_1 + w_2}\right)^2 \cdot {s_{\bar{x}_2}}^2 $$

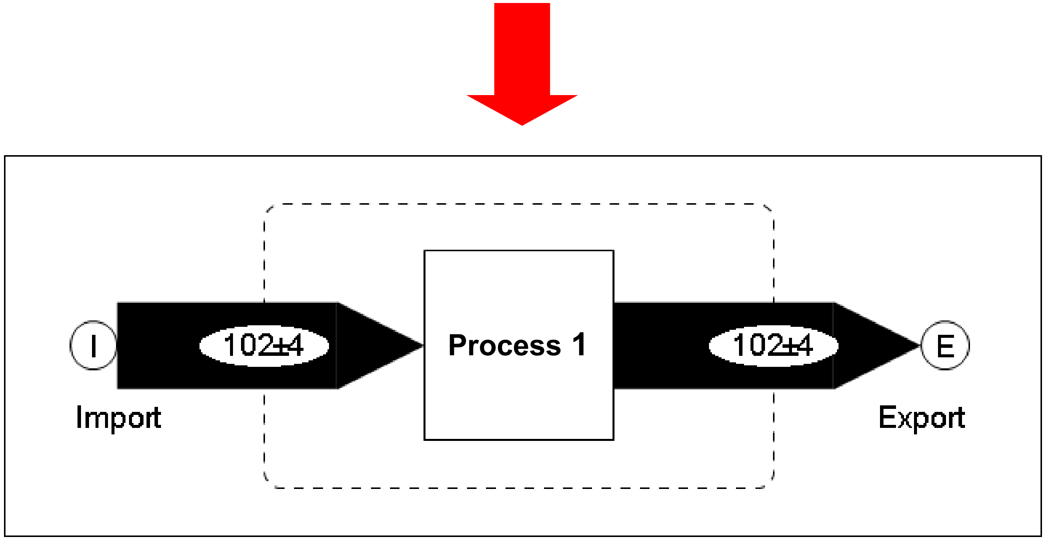
$$ {s_{\bar{x}}}^2 = \left(\frac{w_1}{w_1 + w_2}\right)^2 \cdot \frac{1}{w_1} + \left(\frac{w_2}{w_1 + w_2}\right)^2 \cdot \frac{1}{w_2} $$

$$ {s_{\bar{x}}}^2 = \frac{w_1}{(w_1 + w_2)^2} + \frac{w_2}{(w_1 + w_2)^2} = \frac{w_1 + w_2}{(w_1 + w_2)^2} = \frac{1}{w_1 + w_2} $$

$$ {s_{\bar{x}}}^2 = \frac{1}{\frac{1}{{s_{\bar{x}_1}}^2} + \frac{1}{{s_{\bar{x}_2}}^2}} \quad \text{or} \quad \frac{1}{{s_{\bar{x}}}^2} = \frac{1}{{s_{\bar{x}_1}}^2} + \frac{1}{{s_{\bar{x}_2}}^2} \quad or \quad w_{\bar{x}} = w_1 + w_2 $$

Data Reconciliation (6)



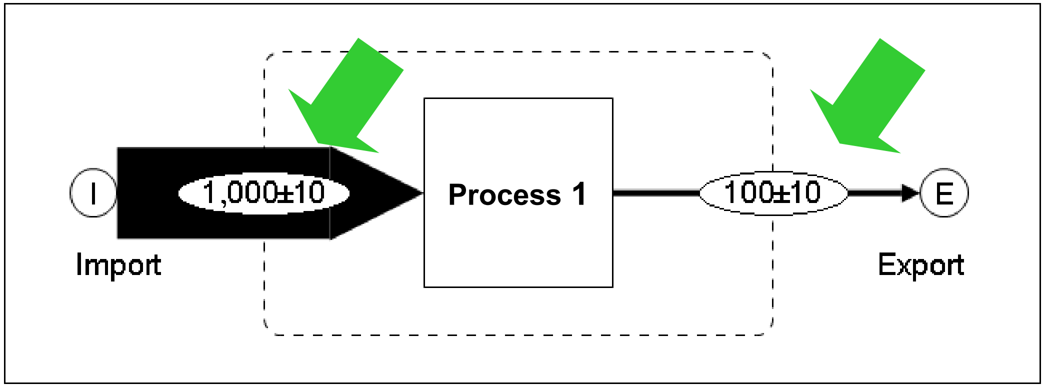


import and export flow have different absolute uncertainties

→ contradiction resolved
→ uncertainty reduced

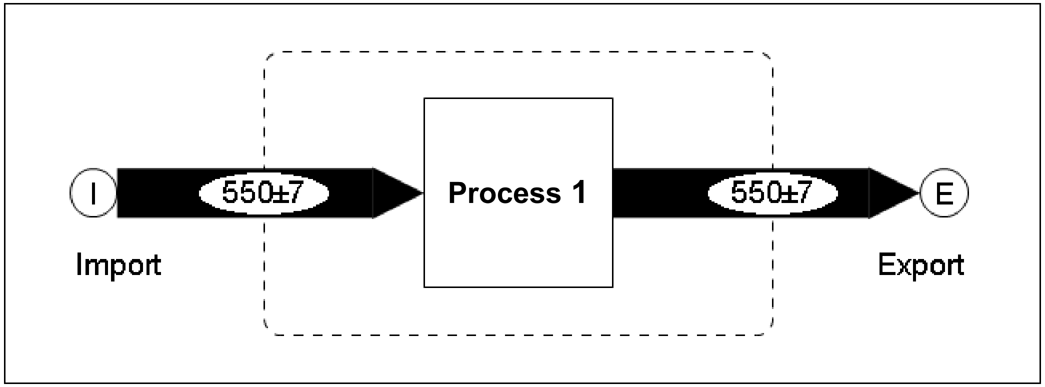
values with higher absolute uncertainty will be reconciled stronger!

Data Reconciliation (7)



same absolute uncertainty

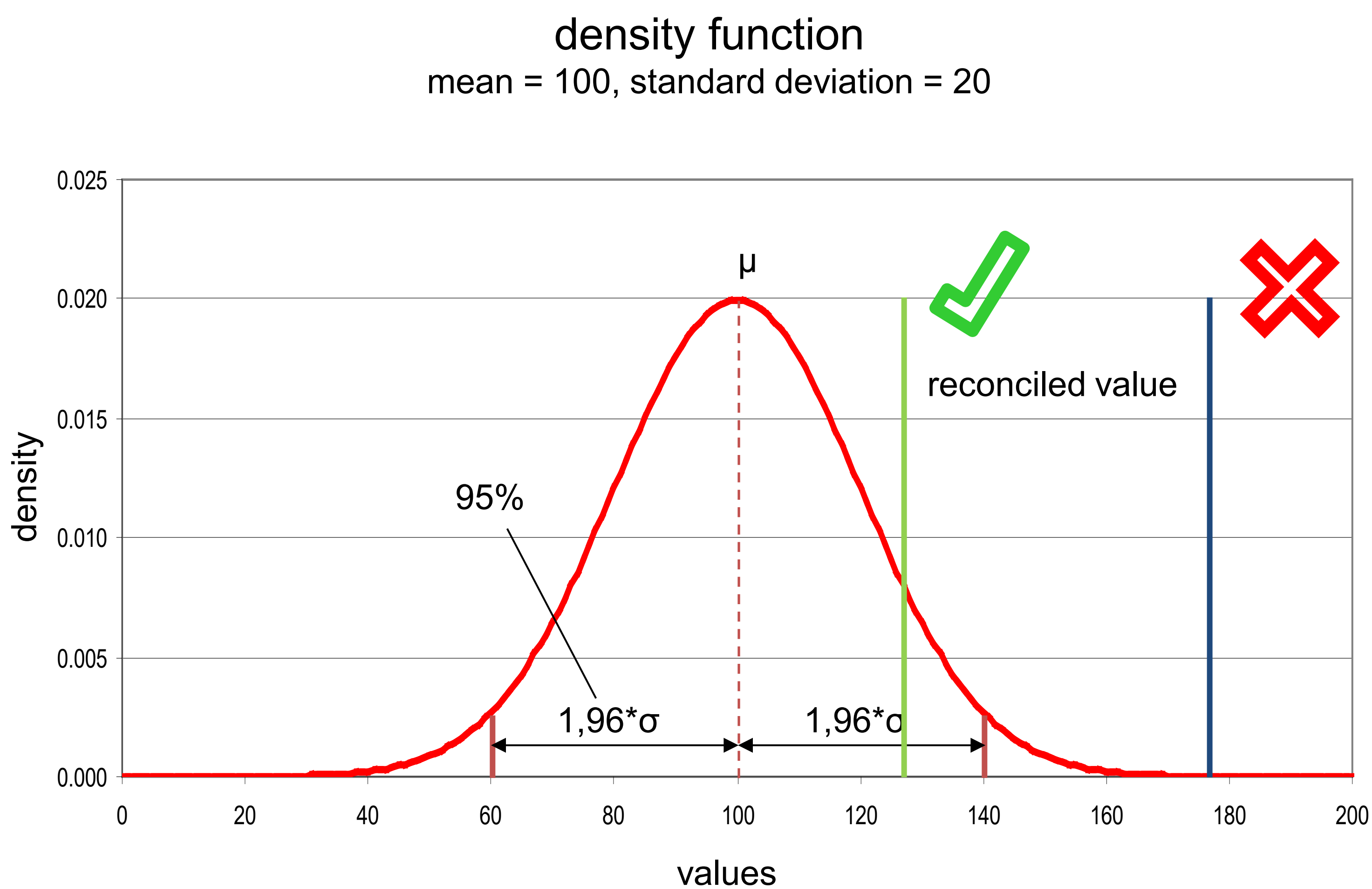
error in import:
1000 instead of 100



→ contradiction resolved
→ uncertainty reduced

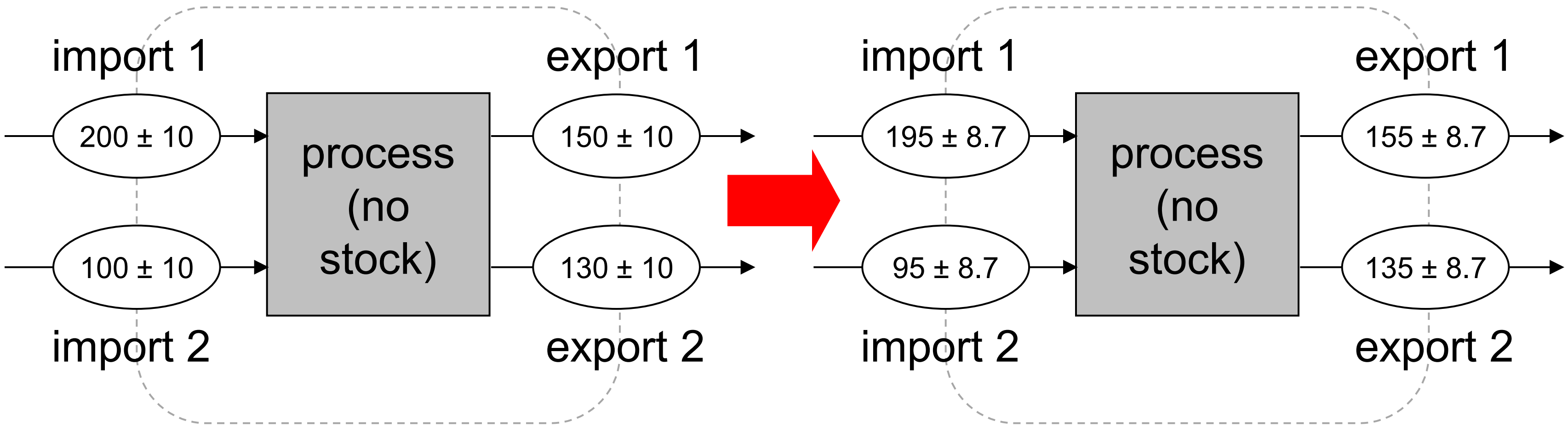
BUT: reconciliation too strong → can be detected by statistical tests

Data Reconciliation (8): Statistical Test



Data Reconciliation (9)

Ex.: all flows have the same standard deviation,
1 balance equation, 0 unknown variables
(= overdetermined system of equations → data reconciliation!)



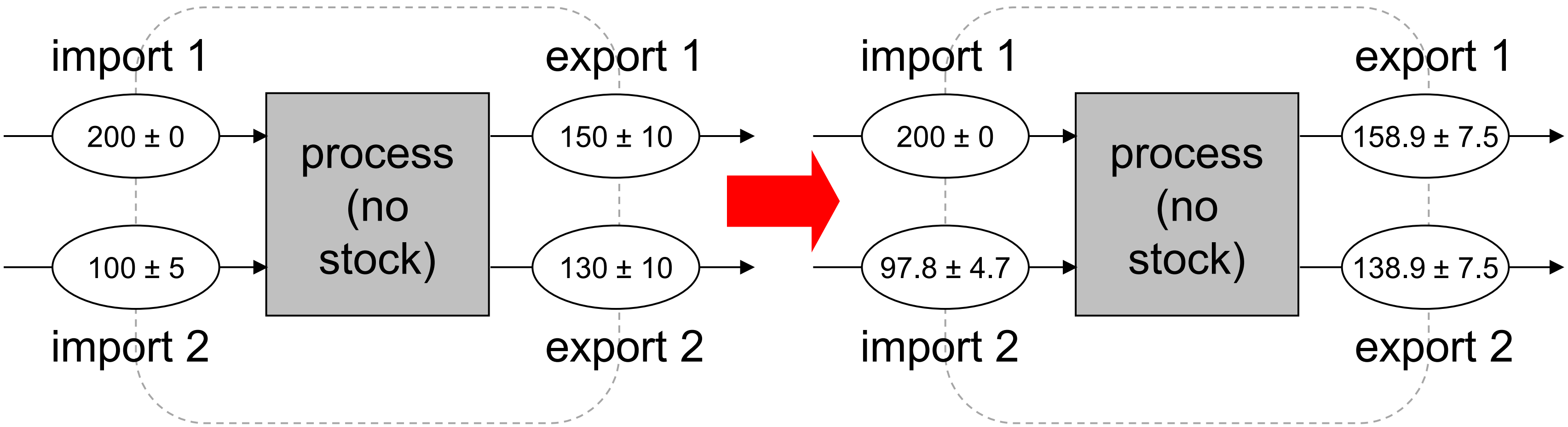
∑ import ≠ ∑ export !

∑ import = ∑ export !

uncertainties are reduced !

Data Reconciliation (10)

Ex.: all flows have the same standard deviation,
1 balance equation, 0 unknown variables
(= overdetermined system of equations → data reconciliation!)

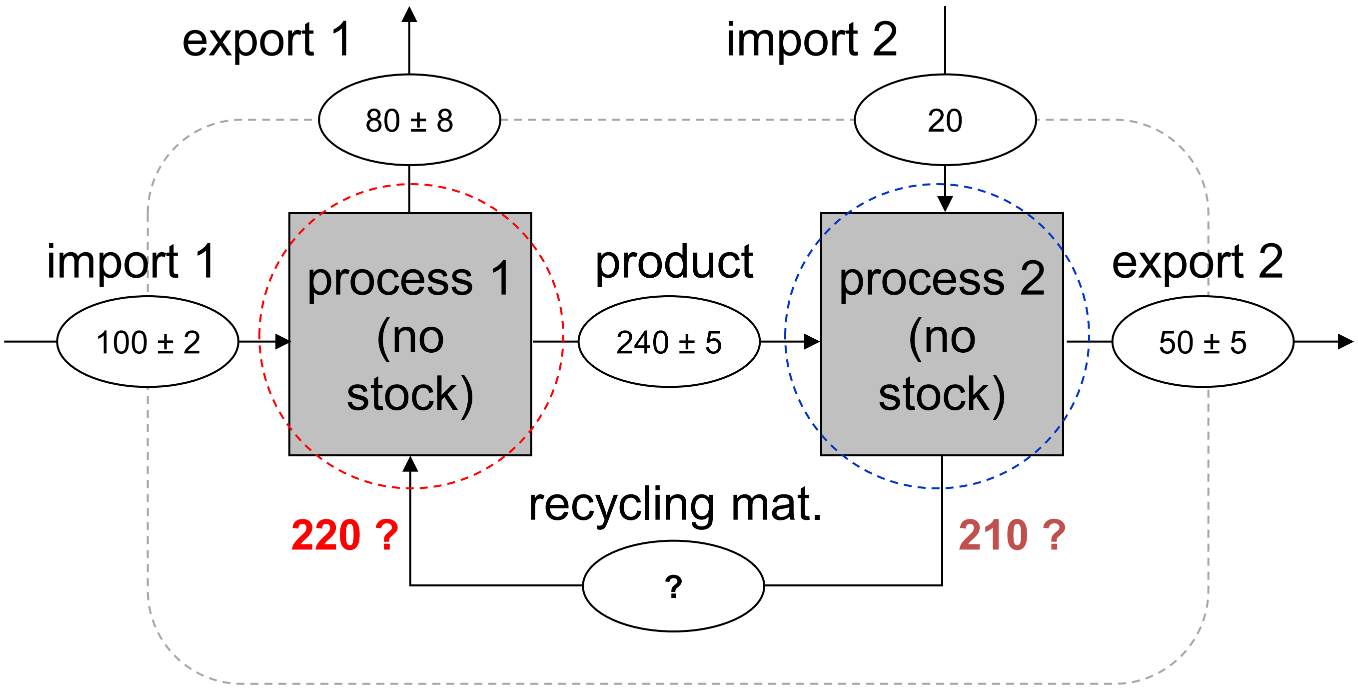


∑ import ≠ ∑ export !

∑ import = ∑ export !

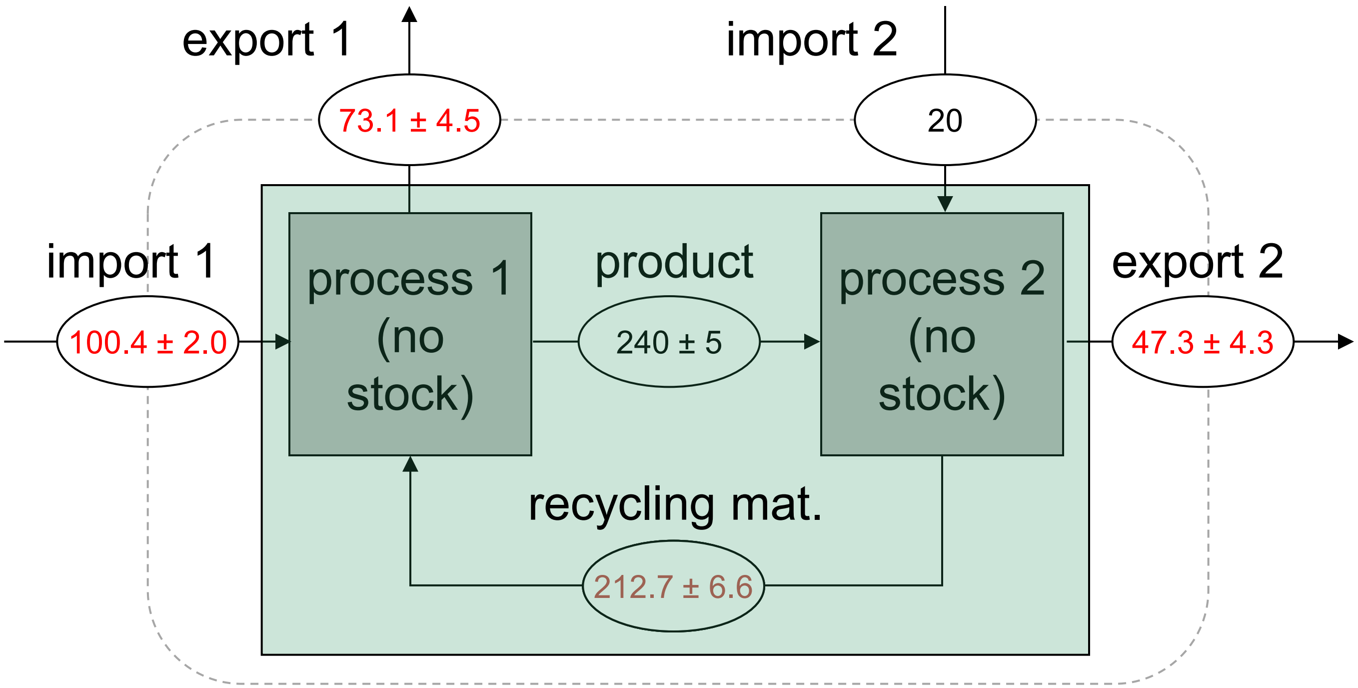
values with higher uncertainty are reconciled stronger!

Data Reconciliation (11)



Data Reconciliation (12)

data reconciliation



calculation
+ error propagation

Error Propagation (1)

C. F. Gauss (1777-1855)

$$Y = f(X_1, X_2, \dots, X_n)$$

X_i … random variable
with mean value $\bar{X}_i$ and variance $var(X_i)$

$$\bar{Y} \approx f(\bar{X}_1, \bar{X}_2, \dots \bar{X}_n)$$

Gauss' Law of Error Propagation:

$$var(Y) \approx \sum_{i=1}^{n} \left(var(X_i) \cdot \left[\frac{\partial Y}{\partial X_i} \right]_{\bar{X}_i}^2 \right)$$

for independent $\bar{X}_i$´s!

Error Propagation (2)

Example:

$$F = f(A, B, C, D) = A + 2B + C \cdot D$$

$$\bar{F} = f(\bar{A}, \bar{B}, \bar{C}, \bar{D}) = \bar{A} + 2\bar{B} + \bar{C} \cdot \bar{D}$$

$$S_F{}^2 = \sum_{P=A\ldots D} \left(\frac{\partial F}{\partial P}\right)^2_{\bar{P}} \cdot S_P{}^2$$

$$\frac{\partial F}{\partial A} = 1, \quad \frac{\partial F}{\partial B} = 2, \quad \frac{\partial F}{\partial C} = D, \quad \frac{\partial F}{\partial D} = C$$

$$S_F{}^2 = 1^2 S_A{}^2 + 2^2 S_B{}^2 + \bar{D}^2 \cdot S_C{}^2 + \bar{C}^2 \cdot S_D{}^2$$

Error Propagation (3)

$$C = f(A, B, \ldots)$$

$$\bar{C} = f(\bar{A}, \bar{B}, \ldots) \qquad {S_C}^2 = \left(\frac{\partial C}{\partial A}\right)^2_{\bar{A},\bar{B},\ldots} \cdot {S_A}^2 + \left(\frac{\partial C}{\partial B}\right)^2_{\bar{A},\bar{B},\ldots} \cdot {S_B}^2 + \ldots$$

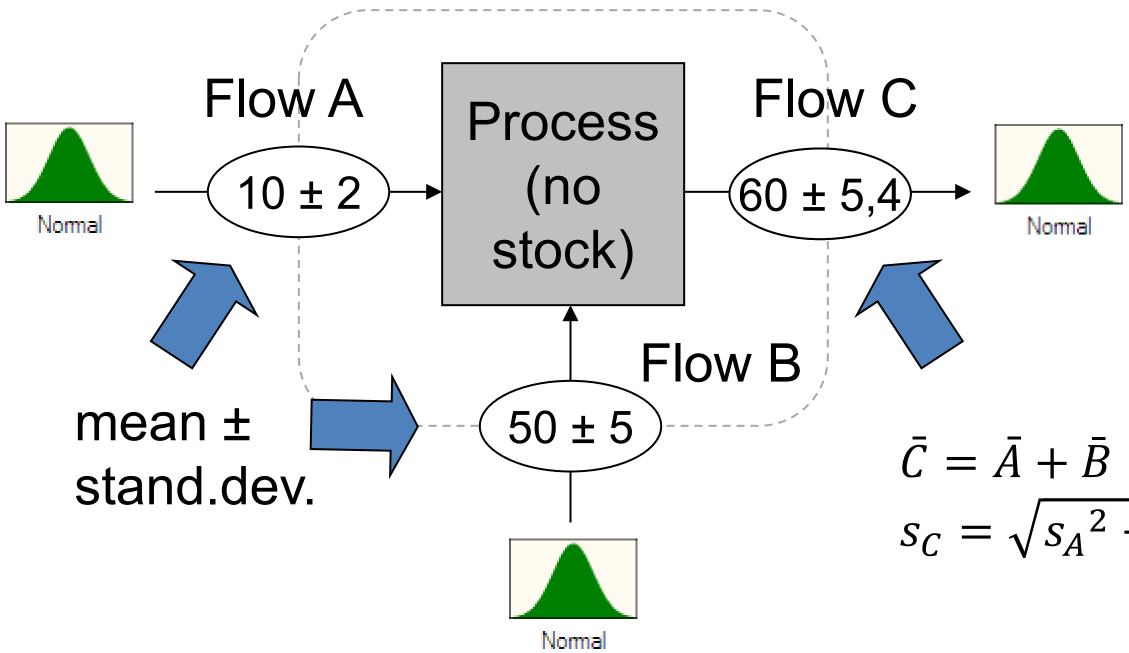
Basic calculation rules:

$$C = A \pm B: \quad {S_C}^2 = {S_A}^2 + {S_B}^2$$

$$C = k \cdot A: \quad {S_C}^2 = k^2 \cdot {S_A}^2$$

$$\left.\begin{aligned} C = A \cdot B: &\quad {S_C}^2 = \bar{B}^2 \cdot {S_A}^2 + \bar{A}^2 \cdot {S_B}^2 \\ C = A/B: &\quad {S_C}^2 = \frac{1}{\bar{B}^2} \cdot {S_A}^2 + \frac{\bar{A}^2}{\bar{B}^4} \cdot {S_B}^2 \end{aligned}\right\} \left(\frac{S_C}{\bar{C}}\right)^2 = \left(\frac{S_A}{\bar{A}}\right)^2 + \left(\frac{S_B}{\bar{B}}\right)^2$$

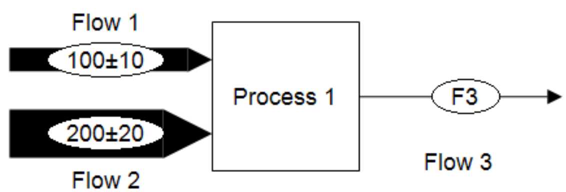
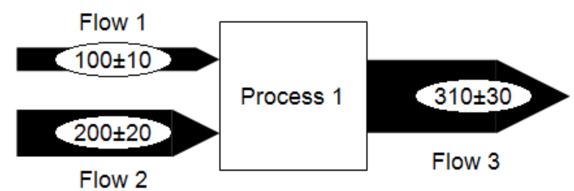
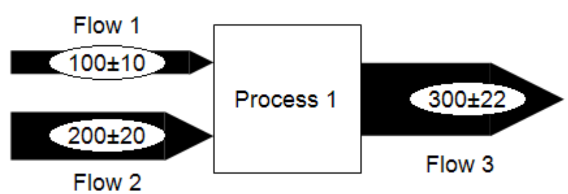
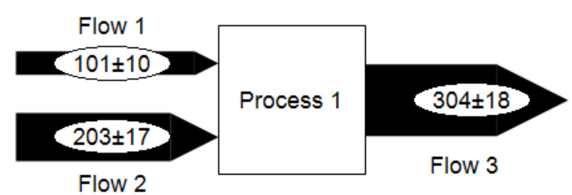
Error Propagation (4)



$$\bar{C} = \bar{A} + \bar{B}$$

$$s_C = \sqrt{s_A{}^2 + s_B{}^2}$$

Error Propagation vs Data Reconciliation

	Error propagation:	Data reconciliation:
Start:		
Requirements:	• exactly determined system of equations (here 1 equation, 1 unknown) • uncertain data given	• over determined system of equations (here 1 equation, 0 unknowns) • uncertain data given
Result:		

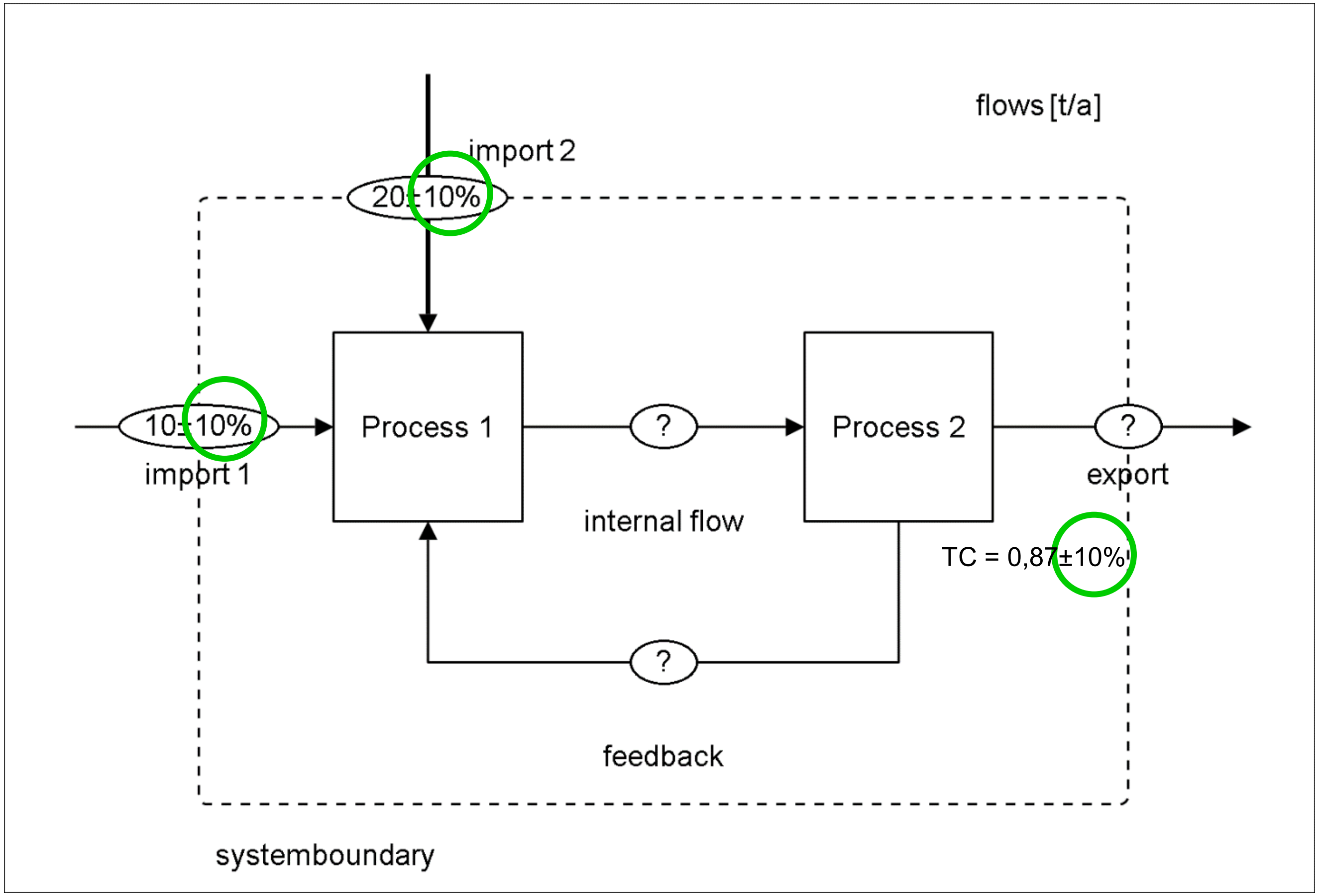
Software for Substance Flow Analysis

STAN 2

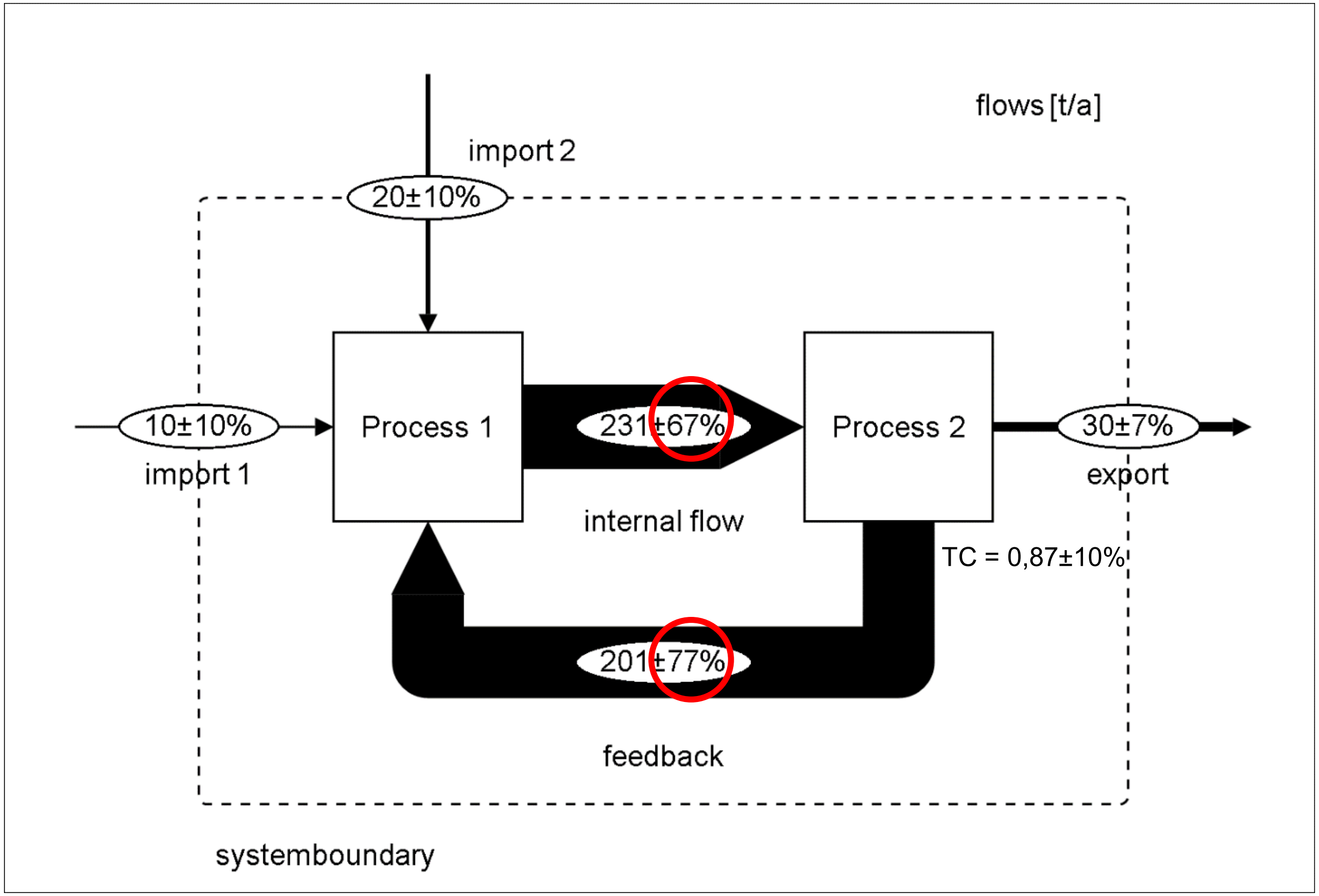
Vienna University of Technology

www.stan2web.net

Limits of Common Sense (1)

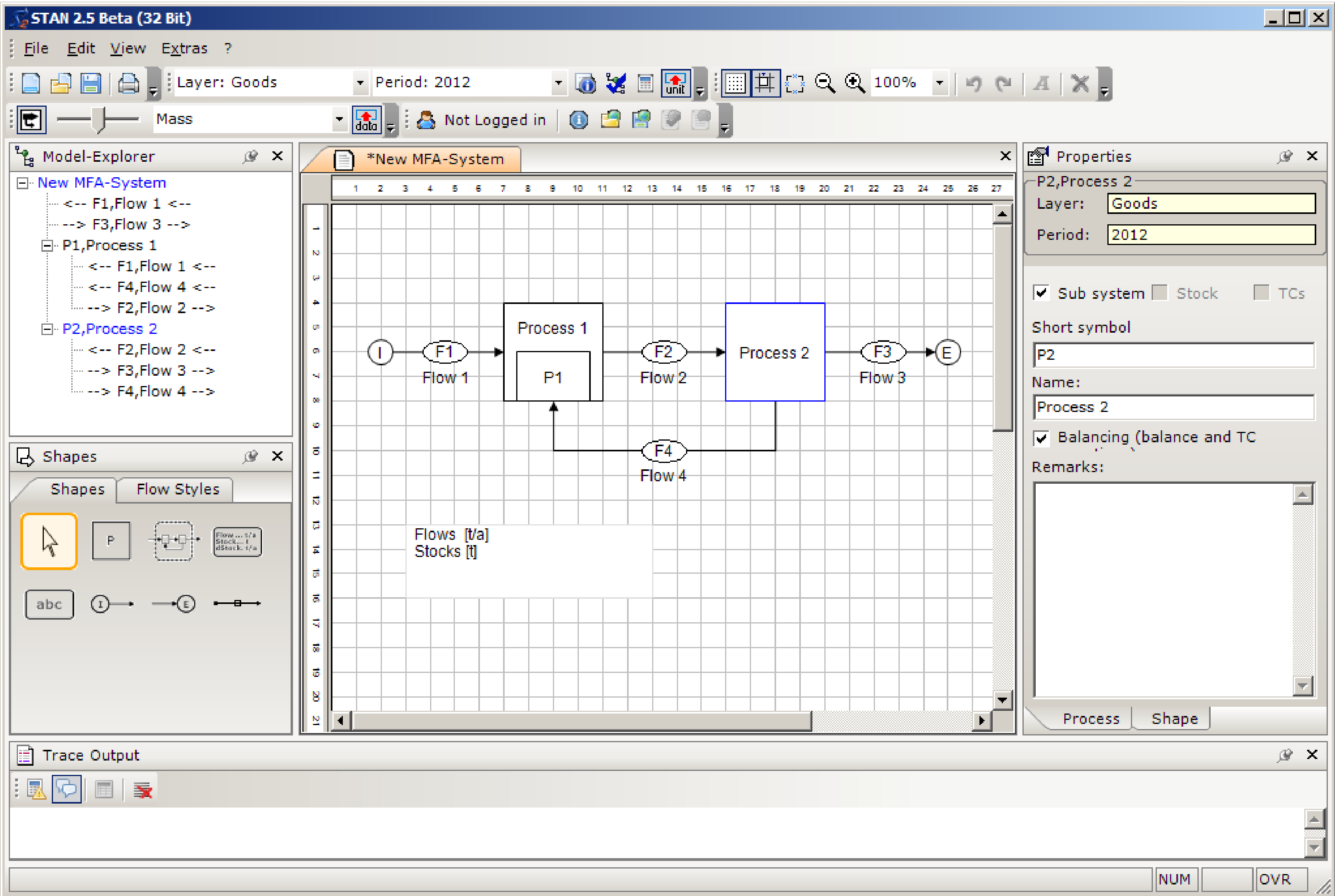


Limits of Common Sense (2)

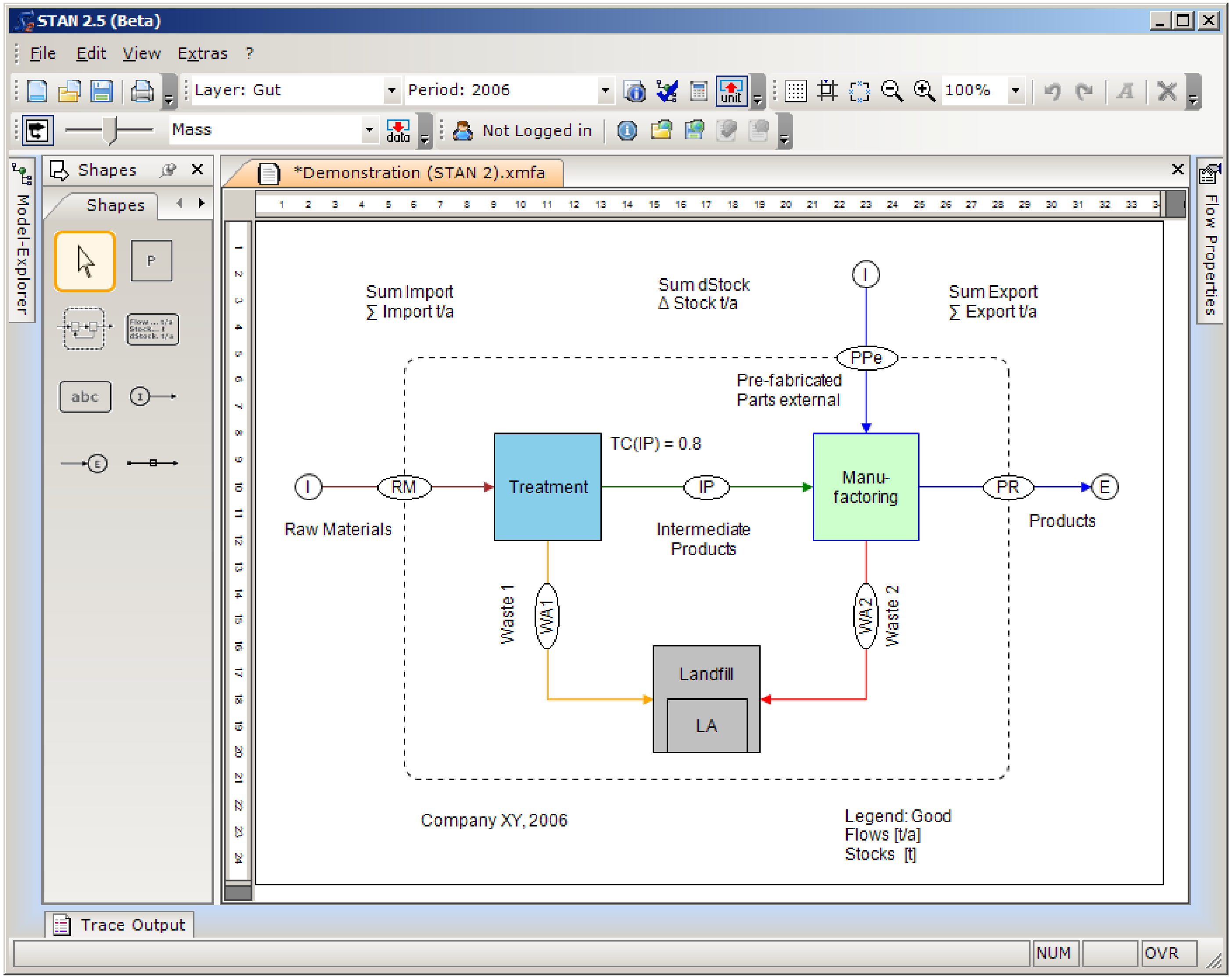


$$FB = \frac{TC \cdot (I1 + I2)}{1 - TC} \qquad (\sigma_{FB})^2 = \left(\frac{\partial FB}{\partial I1}\right)^2 \cdot (\sigma_{I1})^2 + \left(\frac{\partial FB}{\partial I2}\right)^2 \cdot (\sigma_{I2})^2 + \left(\frac{\partial FB}{\partial TC}\right)^2 \cdot (\sigma_{TC})^2$$

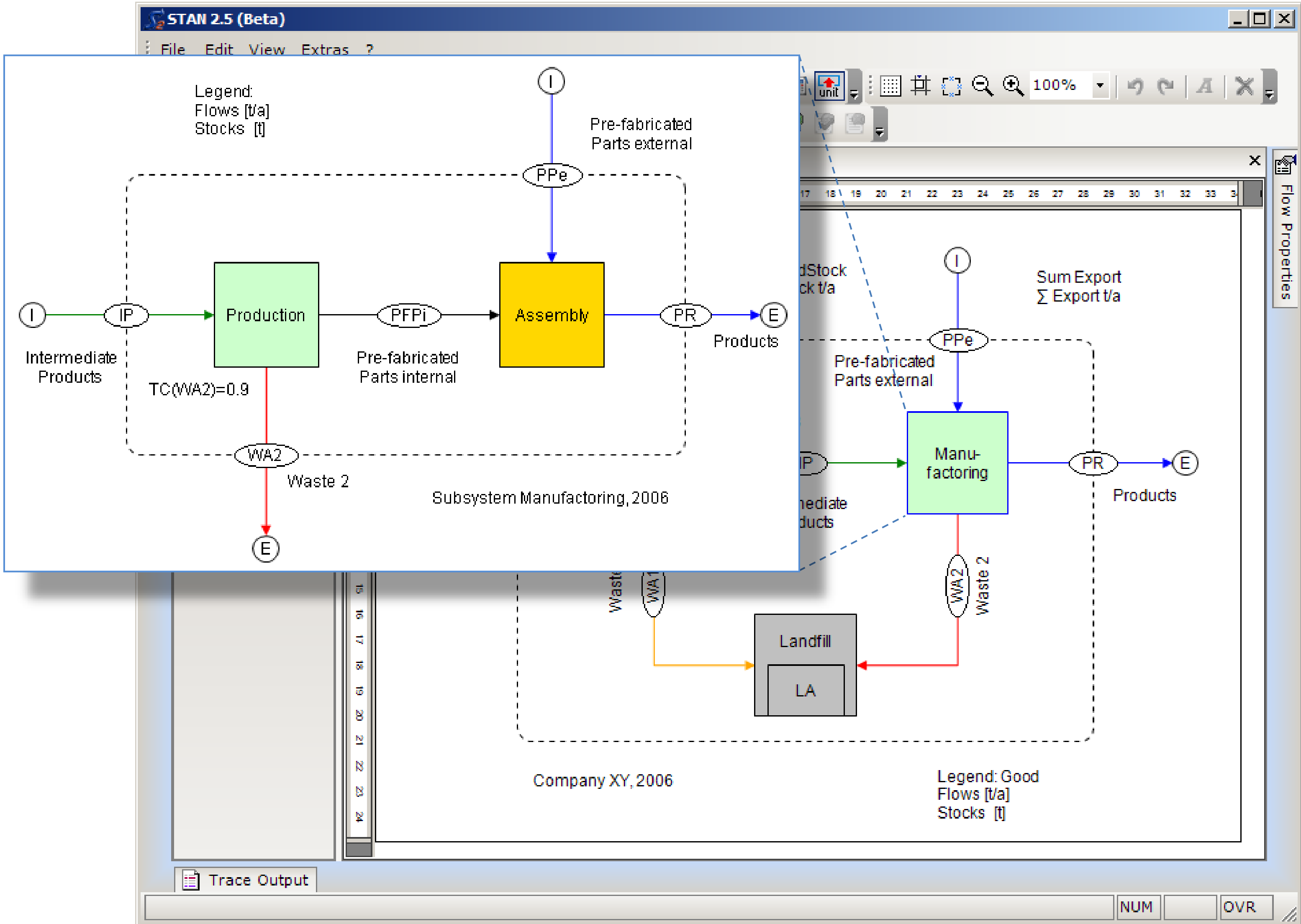
User Interface



Easy Modelling

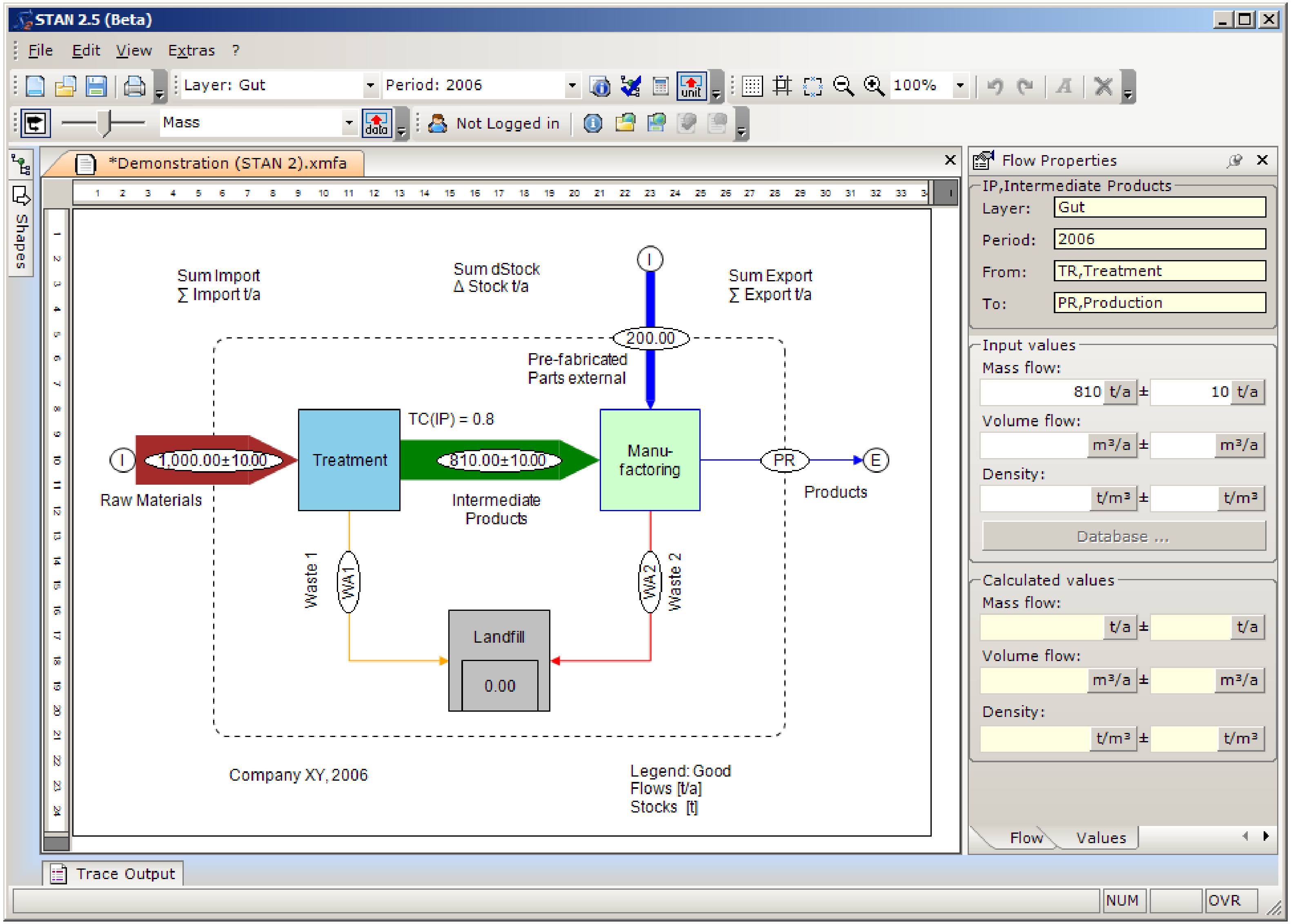


Subsystems

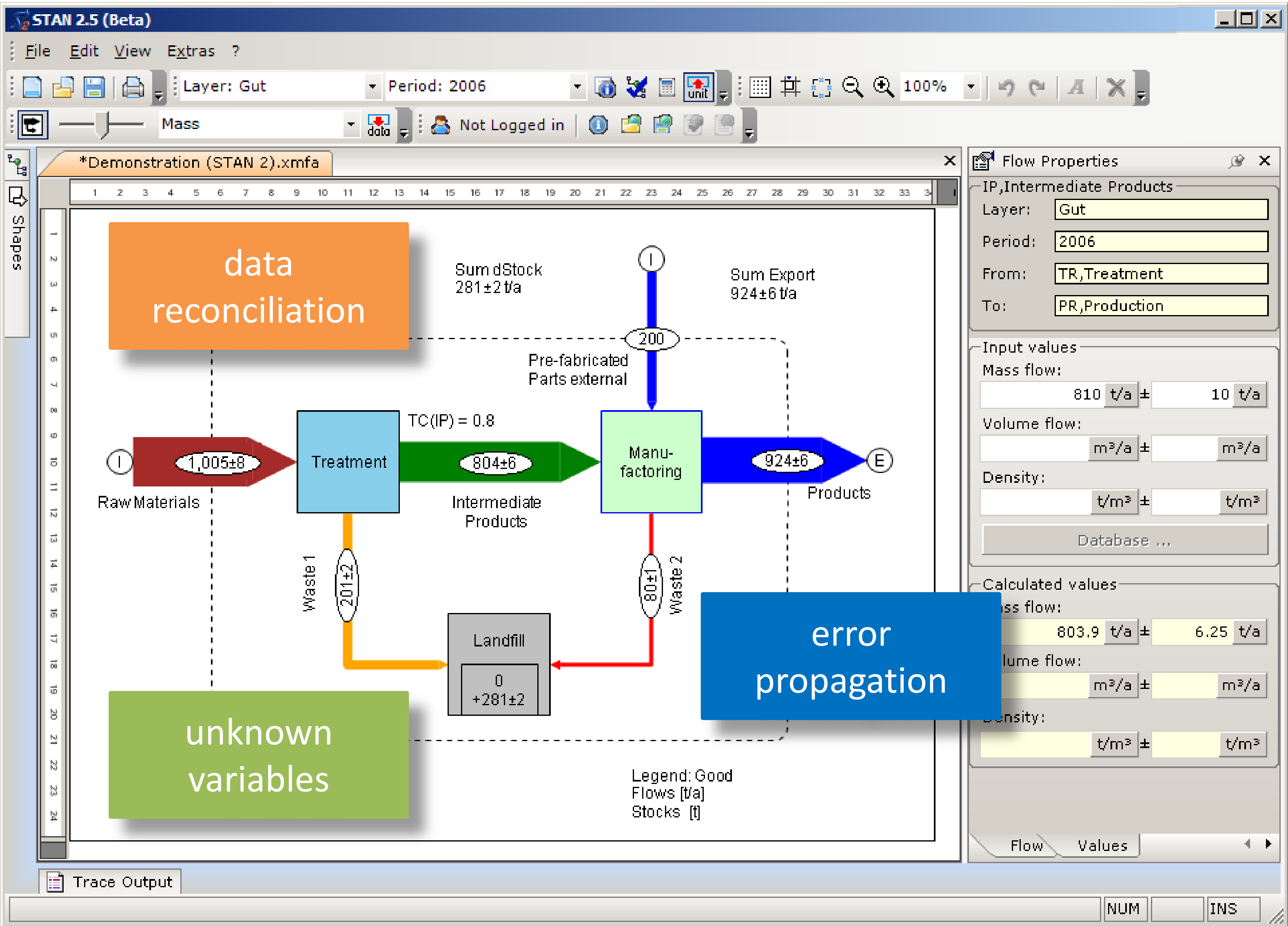


STAN 2.5 (Beta)
File Edit View Extras ?
Legend:
Flows [t/a]
Stocks [t]
I
Pre-fabricated
Parts external
PPe
Production
PFPi
Assembly
PR
E
IP
Intermediate
Products
Products
Pre-fabricated
Parts internal
TC(WA2)=0.9
WA2
Waste 2
Subsystem Manufactoring, 2006
Sum Export
∑ Export t/a
Manu-
factoring
Landfill
LA
WA1
Company XY, 2006
Legend: Good
Flows [t/a]
Stocks [t]
Trace Output
NUM
OVR
Flow Properties

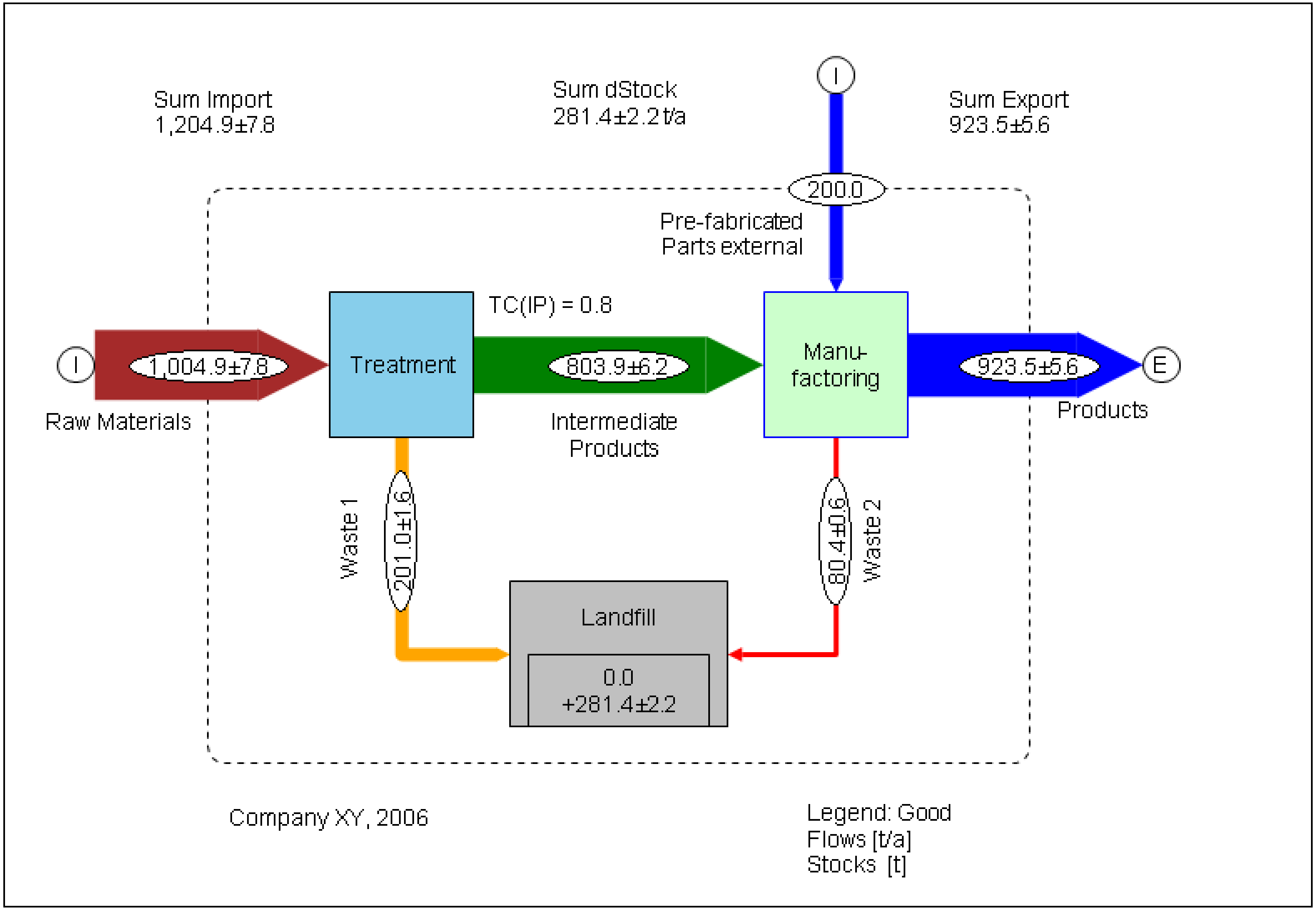
Data Entry


STAN 2.5 (Beta)
File Edit View Extras ?
Layer: Gut
Period: 2006
100%
Mass
Not Logged in
*Demonstration (STAN 2).xmfa
Shapes
Sum Import
∑ Import t/a
Sum dStock
Δ Stock t/a
Sum Export
∑ Export t/a
200.00
Pre-fabricated
Parts external
TC(IP) = 0.8
1,000.00±10.00
Treatment
810.00±10.00
Manu-
facturing
PR
E
Raw Materials
Intermediate
Products
Products
Waste 1
WA1
WA2
Waste 2
Landfill
0.00
Company XY, 2006
Legend: Good
Flows [t/a]
Stocks [t]
Flow Properties
IP,Intermediate Products
Layer: Gut
Period: 2006
From: TR,Treatment
To: PR,Production
Input values
Mass flow:
810 t/a ± 10 t/a
Volume flow:
m³/a ± m³/a
Density:
t/m³ ± t/m³
Database ...
Calculated values
Mass flow:
t/a ± t/a
Volume flow:
m³/a ± m³/a
Density:
t/m³ ± t/m³
Flow
Values
Trace Output
NUM
OVR

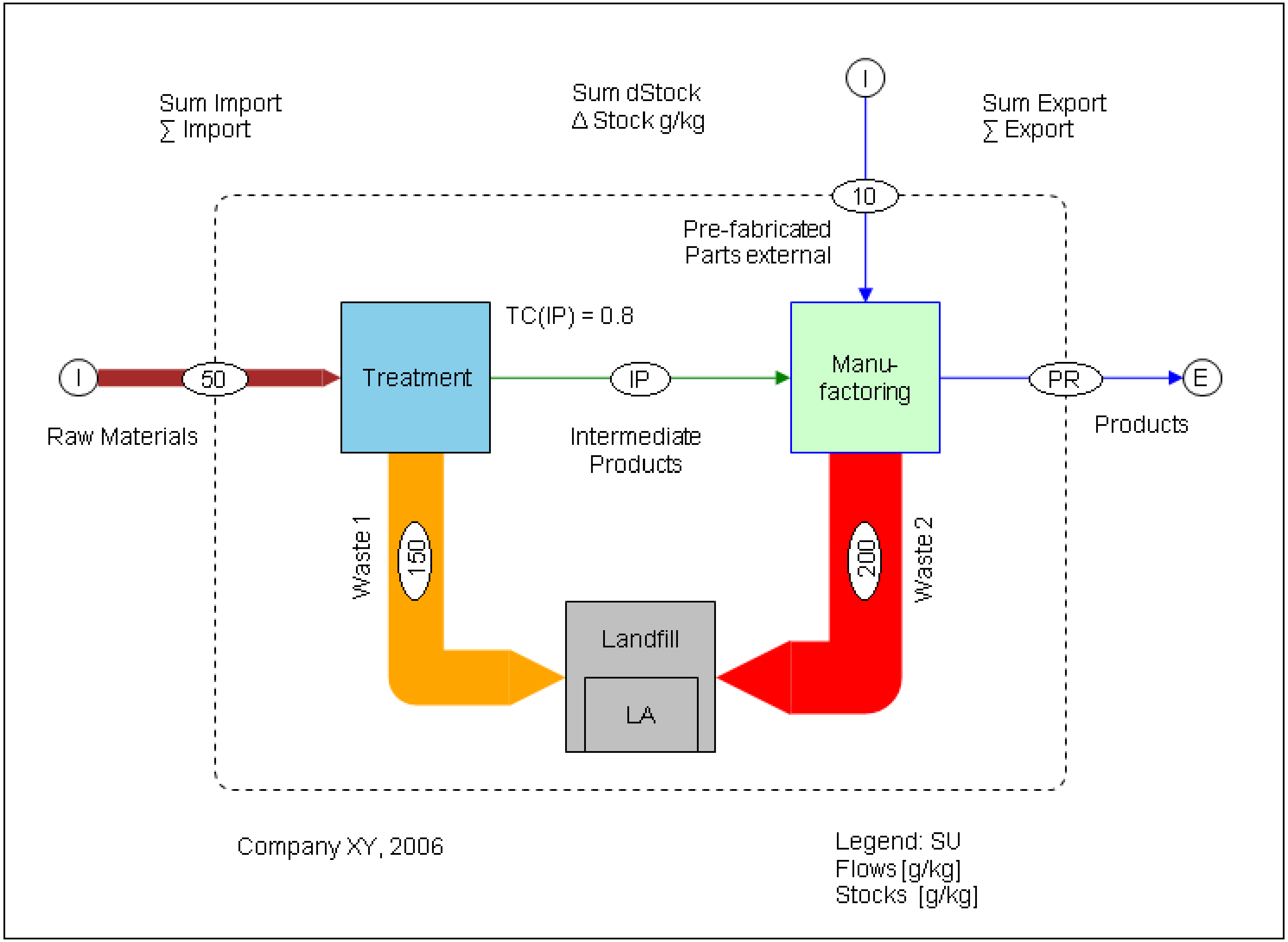
Calculation



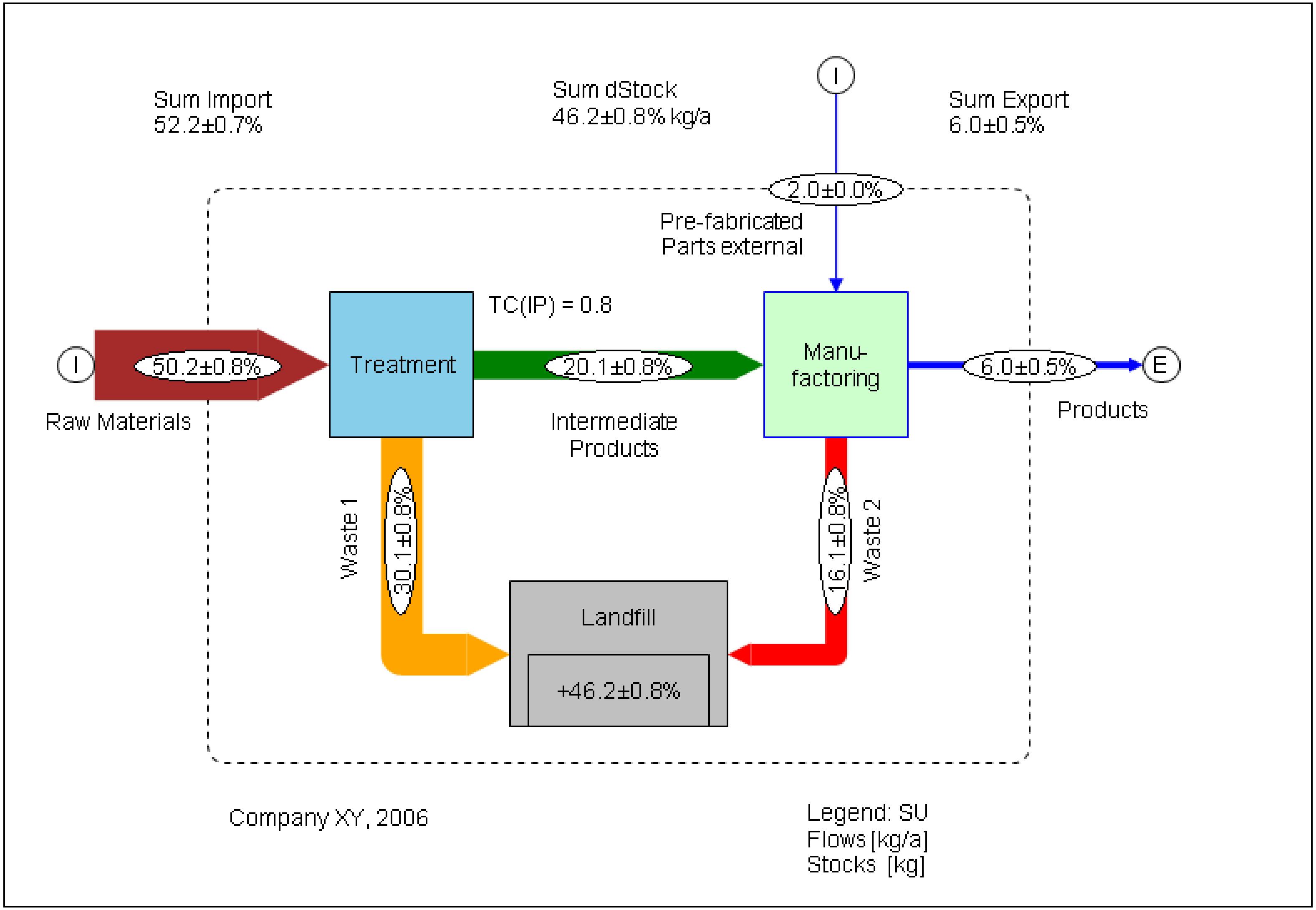
Graph (mass flows of goods)



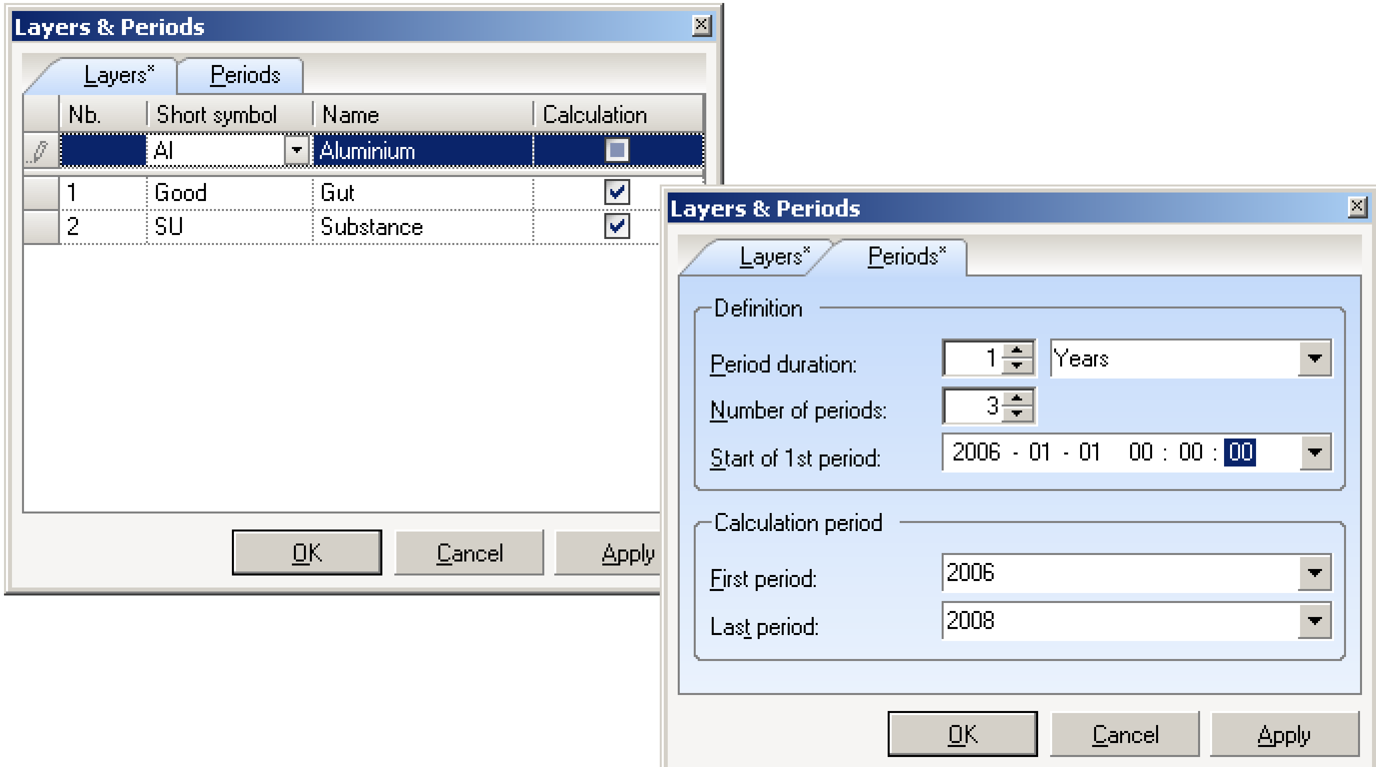
Graph (substance concentrations)

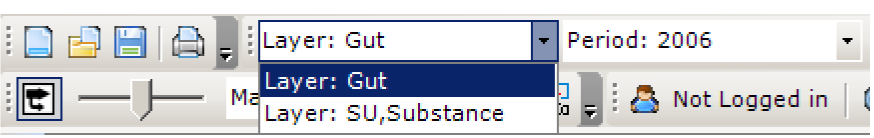


Graph (mass flows of substance)

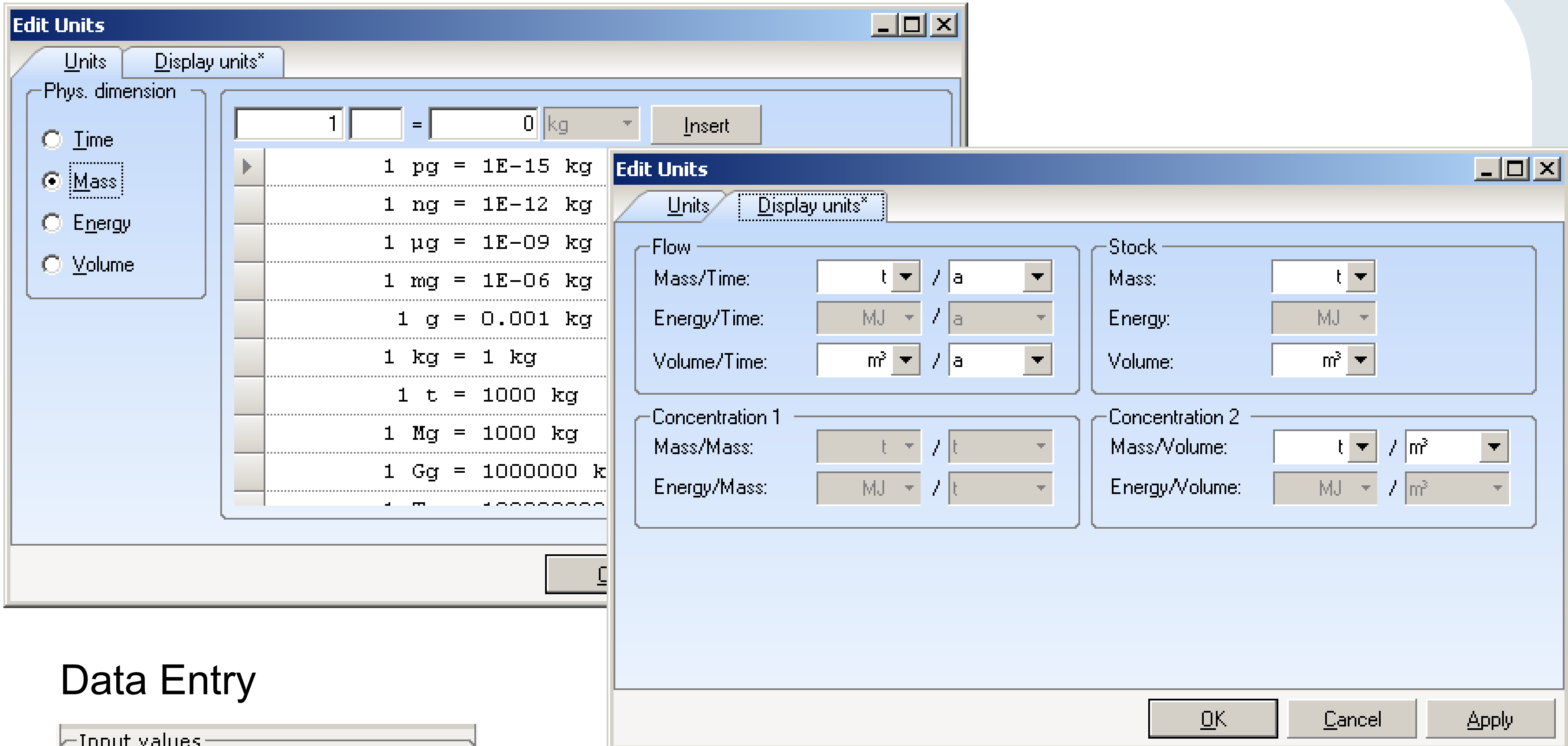


Layers and Periods

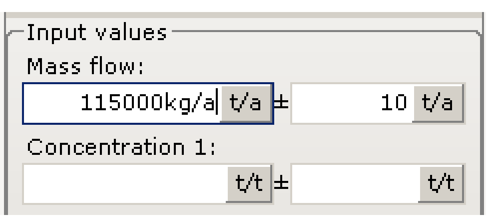




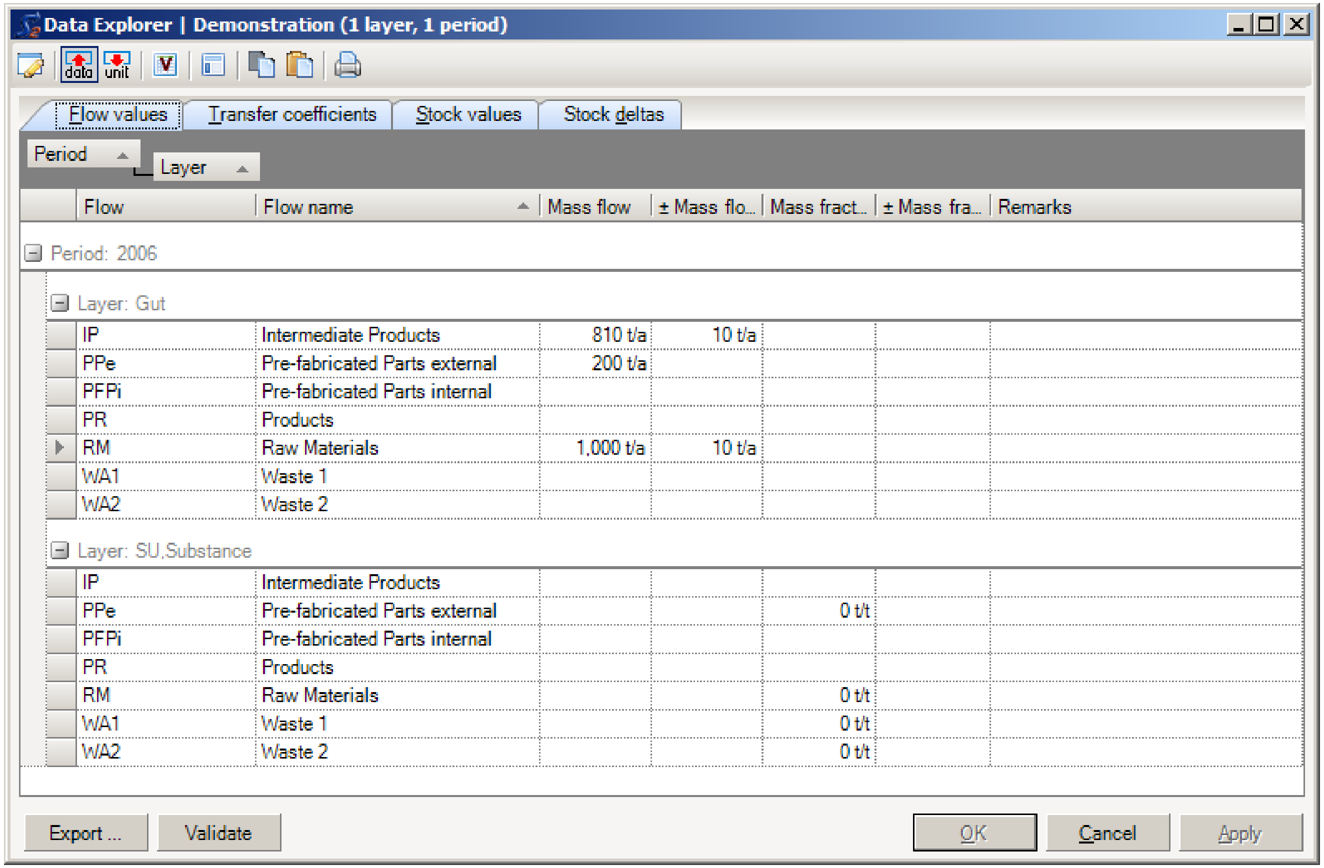
Units and Uncertainties



Data Entry



Data Import/Export



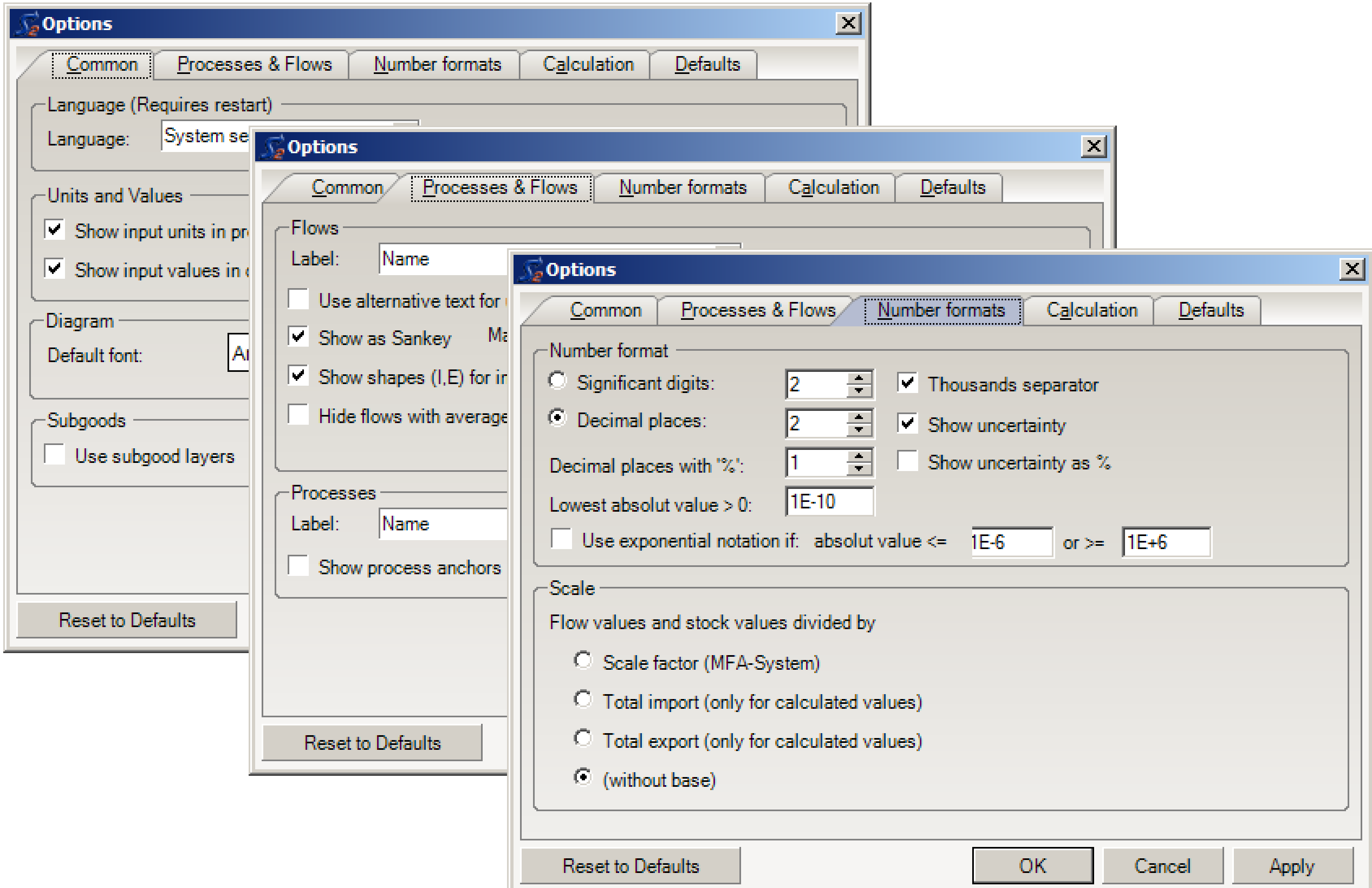
STAN

from/to
Microsoft® Excel

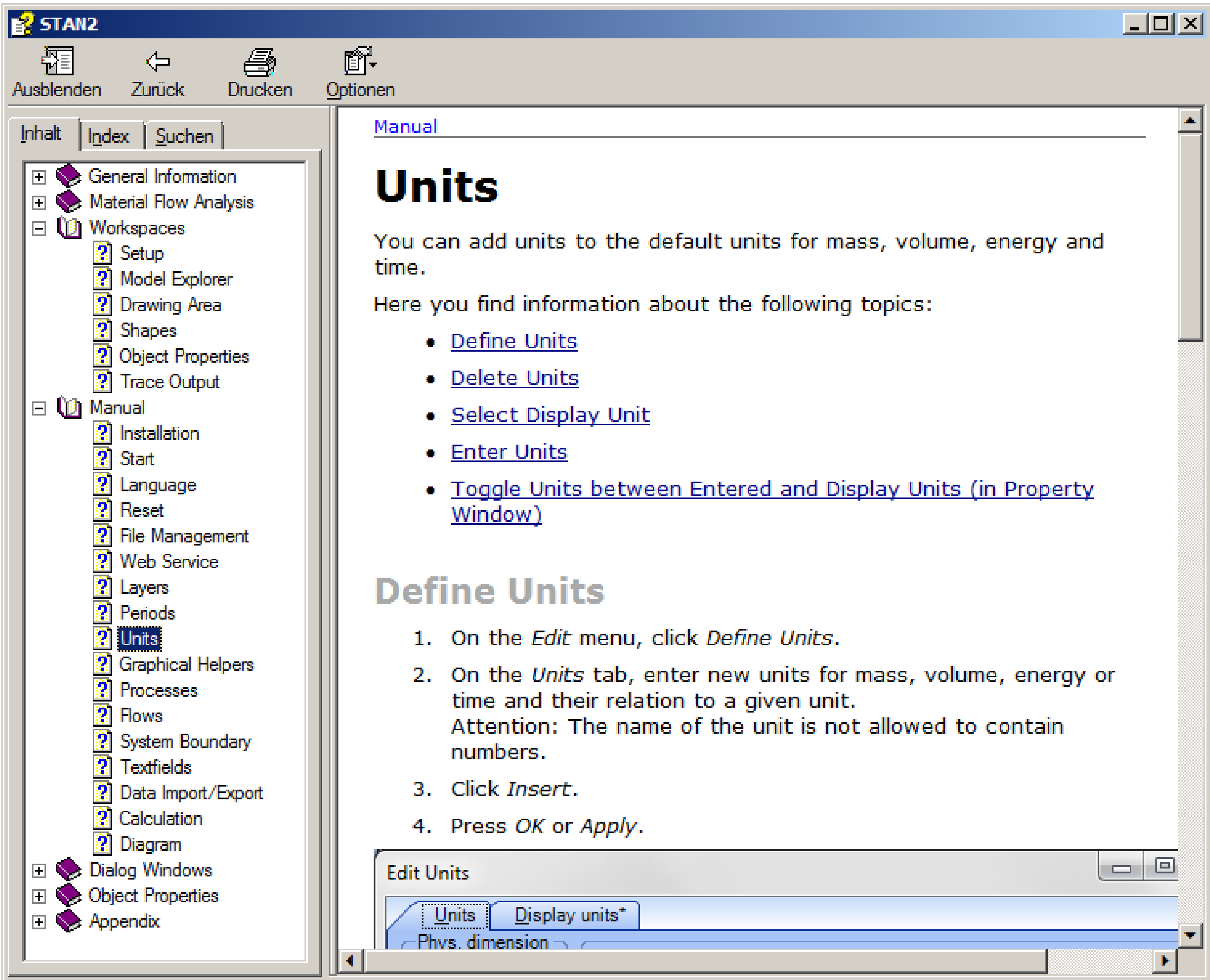
Microsoft Excel

	A	B	C	D	E	F
1	2006	Gut	IP	Intermediate Products	810 t/a	10 t/a
2	2006	Gut	PPe	Pre-fabricated Parts external	200 t/a	

Options

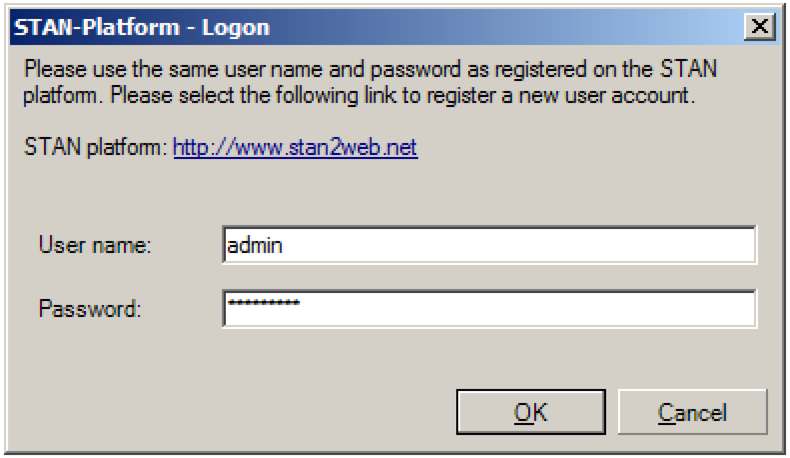

Options
Common
Processes & Flows
Number formats
Calculation
Defaults
Language (Requires restart)
Language:
Units and Values
Diagram
Default font:
Subgoods
Use subgood layers
Reset to Defaults
Options
Flows
Label:
Name
Show as Sankey
Processes
Label:
Name
Show process anchors
Reset to Defaults
Options
Number format
Significant digits:
2
Thousands separator
Decimal places:
2
Show uncertainty
Decimal places with '%':
1
Show uncertainty as %
Lowest absolut value > 0:
1E-10
Use exponential notation if: absolut value <=
1E-6
or >=
1E+6
Scale
Flow values and stock values divided by
Scale factor (MFA-System)
Total import (only for calculated values)
Total export (only for calculated values)
(without base)
Reset to Defaults
OK
Cancel
Apply

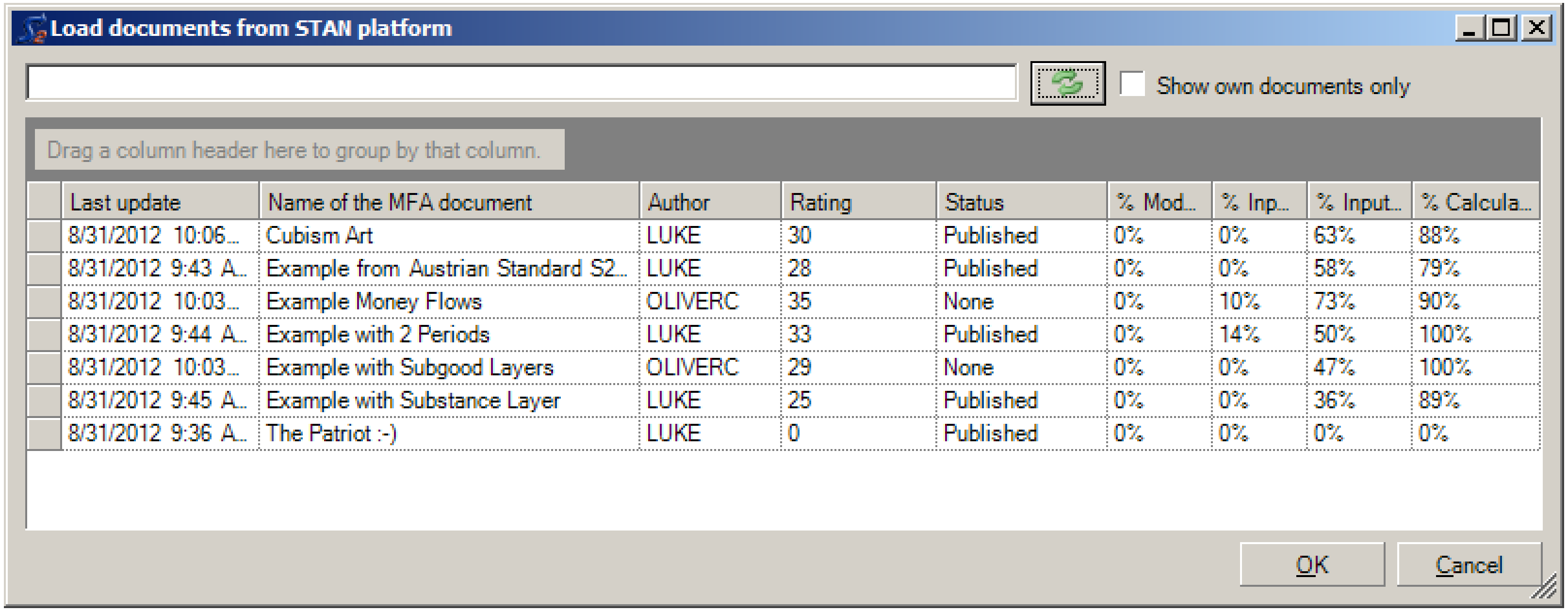
Help (F1)


STAN2
Ausblenden
Zurück
Drucken
Optionen
Inhalt
Index
Suchen
General Information
Material Flow Analysis
Workspaces
Setup
Model Explorer
Drawing Area
Shapes
Object Properties
Trace Output
Manual
Installation
Start
Language
Reset
File Management
Web Service
Layers
Periods
Units
Graphical Helpers
Processes
Flows
System Boundary
Textfields
Data Import/Export
Calculation
Diagram
Dialog Windows
Object Properties
Appendix
Manual
Units
You can add units to the default units for mass, volume, energy and time.
Here you find information about the following topics:
• Define Units
• Delete Units
• Select Display Unit
• Enter Units
• Toggle Units between Entered and Display Units (in Property Window)
Define Units
1. On the Edit menu, click Define Units.
2. On the Units tab, enter new units for mass, volume, energy or time and their relation to a given unit.
Attention: The name of the unit is not allowed to contain numbers.
3. Click Insert.
4. Press OK or Apply.
Edit Units
Units
Display units*
Phys. dimension

Web Interface

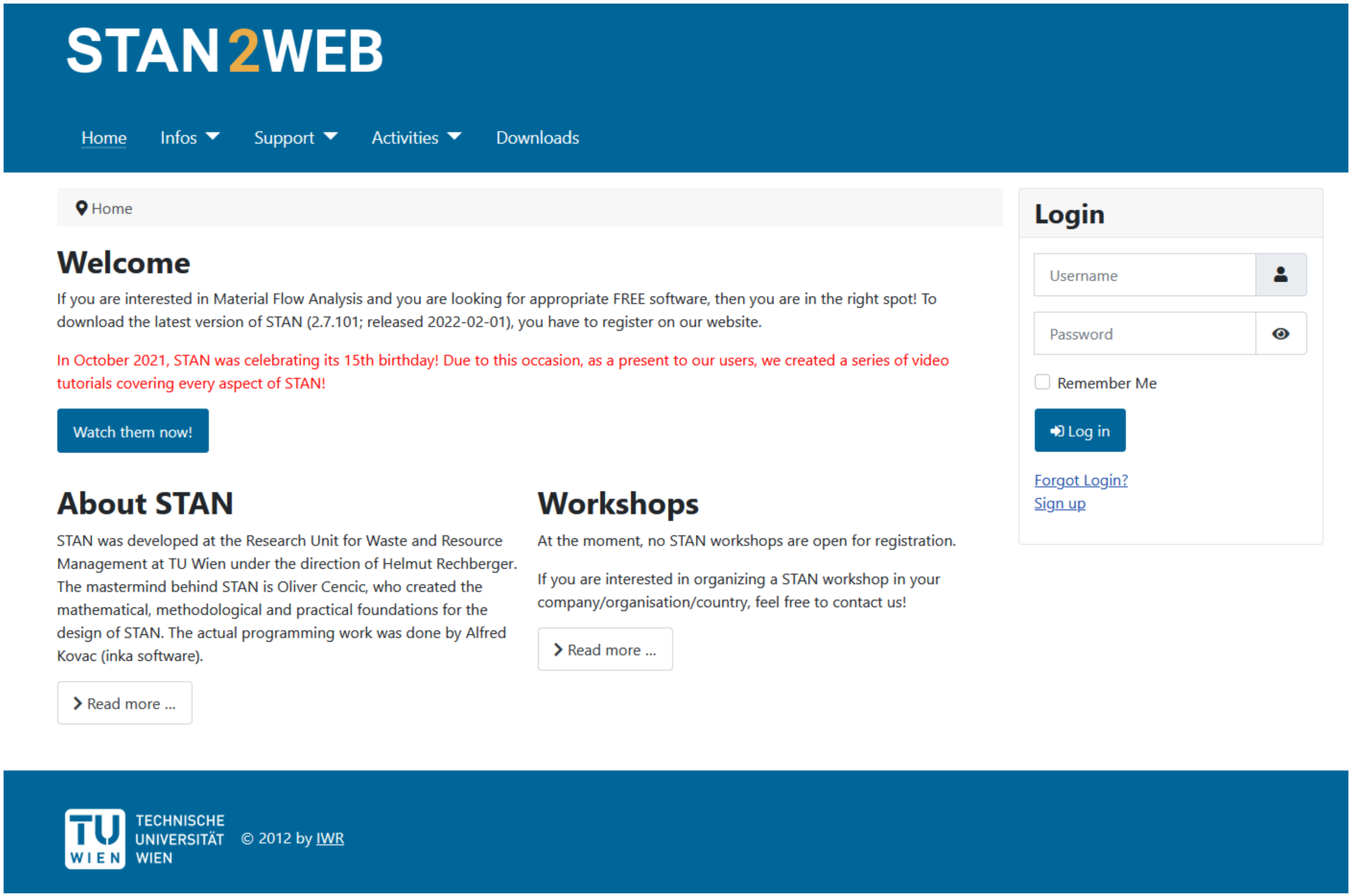
Direct Login into
Web Database





Last update	Name of the MFA document	Author	Rating	Status	% Mod...	% Inp...	% Input...	% Calcula...
8/31/2012 10:06...	Cubism Art	LUKE	30	Published	0%	0%	63%	88%
8/31/2012 9:43 A...	Example from Austrian Standard S2...	LUKE	28	Published	0%	0%	58%	79%
8/31/2012 10:03...	Example Money Flows	OLIVERC	35	None	0%	10%	73%	90%
8/31/2012 9:44 A...	Example with 2 Periods	LUKE	33	Published	0%	14%	50%	100%
8/31/2012 10:03...	Example with Subgood Layers	OLIVERC	29	None	0%	0%	47%	100%
8/31/2012 9:45 A...	Example with Substance Layer	LUKE	25	Published	0%	0%	36%	89%
8/31/2012 9:36 A...	The Patriot :-)	LUKE	0	Published	0%	0%	0%	0%

Website: stan2web.net


STAN2WEB

Home | Infos | Support | Activities | Downloads

Home

Welcome

If you are interested in Material Flow Analysis and you are looking for appropriate FREE software, then you are in the right spot! To download the latest version of STAN (2.7.101; released 2022-02-01), you have to register on our website.

In October 2021, STAN was celebrating its 15th birthday! Due to this occasion, as a present to our users, we created a series of video tutorials covering every aspect of STAN!

Watch them now!

About STAN

STAN was developed at the Research Unit for Waste and Resource Management at TU Wien under the direction of Helmut Rechberger. The mastermind behind STAN is Oliver Cencic, who created the mathematical, methodological and practical foundations for the design of STAN. The actual programming work was done by Alfred Kovac (inka software).

Read more ...

Workshops

At the moment, no STAN workshops are open for registration.

If you are interested in organizing a STAN workshop in your company/organisation/country, feel free to contact us!

Read more ...

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Password

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Thanks for your attention!